REDUCTION OF THE MINIMAL-101 ROOT-4 ODD PERFECT NUMBER BRANCH TO THE SINGLE FACTOR $\Phi_{31}(269)$

ARGUS OPEN RESEARCH PROJECT

Abstract. In the factor-chain tree search for an odd perfect number $N = q^k m^2$ on the minimal-101, special-prime-5, α_1-root-4 branch, we replay 6,756,355 strict-suffix prefixes from a certified 2,054-entry factor database and prove that every branch terminates except at the single mandatory cyclotomic value $\Phi_{31}(269) = \sigma(269^{30}) = (269^{31} - 1)/268$, a 73-digit/96-digit cofactor input outside the pre-certified factor table.

1. Main Factor-Chain Reduction

Proposition 1.1. *Starting from the certified* 2,054*-entry prime-power divisor-sum database on the root-*4 *branch, exhaustive strict-suffix propagation across* 6,756,355 *prefixes closes every divisor-sum branch save for the single unfactored cyclotomic value* $\Phi_{31}(269) = \sigma(269^{30})$.

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